

Online Retail Sales & Profitability Analysis

TOOLS USED

- **Power BI** (Interactive Dashboards & Visuals)
- **Power Query** (Data ETL & Transformation)
- **DAX** (Data Analysis Expressions & Measures)

DATASET OVERVIEW

- **Sample Superstore Dataset**
- **Total Orders Analyzed:** 9,994 Orders
- **Timeframe:** 2014–2017 Historical Sales Data

Total Sales

\$2.30M

Total Profit

\$286.40K

Profit Margin (%)

12.47%

Total Orders

5K

Business Problem & Objectives

BUSINESS PROBLEM

The retail company has thousands of orders across multiple regions, categories, products, and customer segments. Management lacks visibility into which areas drive revenue and profitability and which areas need improvement.

KEY BUSINESS QUESTIONS

- How has sales performance changed over time?
- Which regions generate the highest sales?
- Which product categories contribute the most profit?
- Which customer segments drive revenue?
- Which sub-categories are reducing profitability?
- Which products generate the highest sales?

PROJECT OBJECTIVE

Analyze historical sales data from 2014–2017 to identify:

- Revenue trends
- Profit drivers
- Customer purchasing behavior
- High-performing products
- Loss-making areas
- Growth opportunities

Executive Dashboard Overview

Total Sales

\$2.30M

Total Profit

\$286.40K

Profit Margin (%)

12.47%

Total Orders

5K

Total Customers

793

Avg Order Value

\$458.61

Sales Performance Analysis



KEY INSIGHTS

- Sales generally increased from 2014 to 2017.
- Sales went up and down during certain periods, showing seasonal buying patterns.
- The highest sales were recorded in 2017.
- Sales growth became stronger in the final year.
- Overall demand remained healthy throughout the period.
-

KEY FINDINGS

- Sales growth confirms increasing market demand.
- Seasonal patterns can help forecast inventory requirements effectively.
- High-performing periods can be leveraged for targeted promotions and marketing campaigns.

Regional Performance Analysis



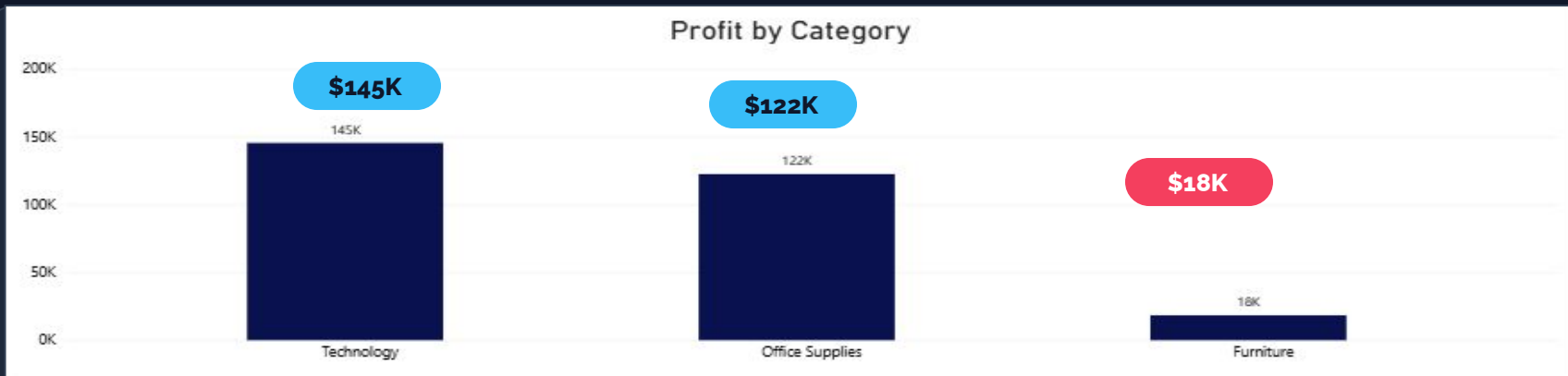
KEY INSIGHTS

- The West region generated the highest sales, reaching **\$0.73M**.
- The East region followed closely with **\$0.68M** in sales.
- The Central region contributed **\$0.50M** in revenue.
- The South region recorded the lowest sales at **\$0.39M**.
- Sales performance differs significantly across regions.
-

KEY FINDINGS

- The West and East regions are the strongest contributors to overall revenue.
- The South region has the greatest opportunity for future growth.
- Regional sales differences suggest variations in customer demand and market reach.
- Targeted marketing and business expansion efforts can help improve performance in lower-performing regions.
-

Category Profitability Analysis



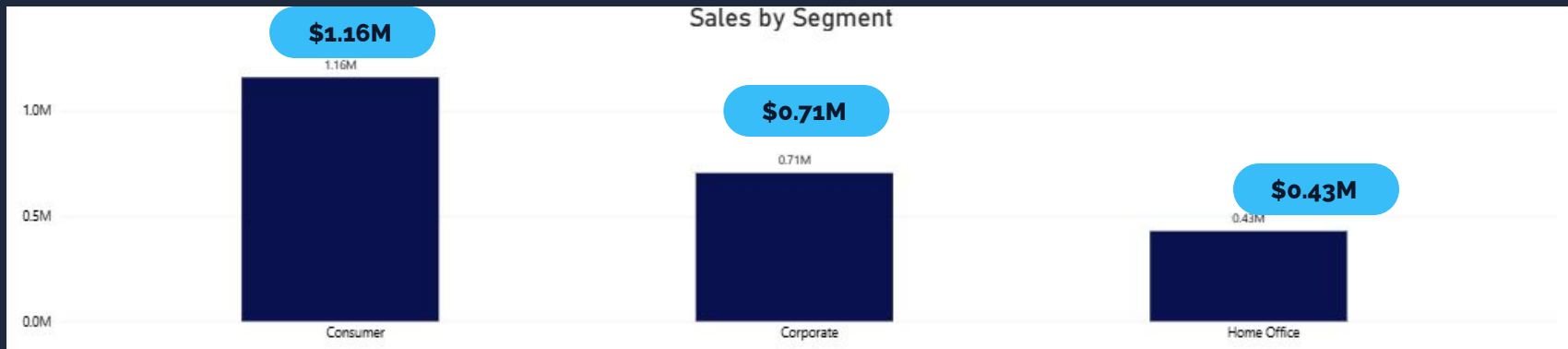
KEY INSIGHTS

- Technology was the most profitable category, generating **\$145K** in profit.
- Office Supplies also performed well, contributing **\$122K** in profit.
- Furniture generated only **\$18K** in profit, making it the least profitable category.
- Most of the company's profit comes from Technology and Office Supplies.

KEY FINDINGS

- Technology and Office Supplies are the main profit drivers for the business.
- Furniture sales may be strong, but its profit contribution is relatively low.
- Profitability is heavily concentrated in two categories.
- The business should investigate factors affecting Furniture profitability, such as discounts, pricing, or costs.

Customer Segment Analysis



KEY INSIGHTS

- The Consumer segment generated the highest sales, contributing **\$1.16M** in revenue.
- The Corporate segment generated **\$0.71M** in sales.
- The Home Office segment contributed **\$0.43M** in revenue.
- The Consumer segment is the largest customer group for the business.

KEY FINDINGS

- Consumer customers are the primary source of revenue.
- Corporate customers also contribute a significant share of sales.
- Home Office is the smallest segment but still represents a valuable customer group.
- Revenue is largely driven by Consumer customers, making this segment important for future growth strategies.

Sub-Category Profitability Analysis



KEY INSIGHTS

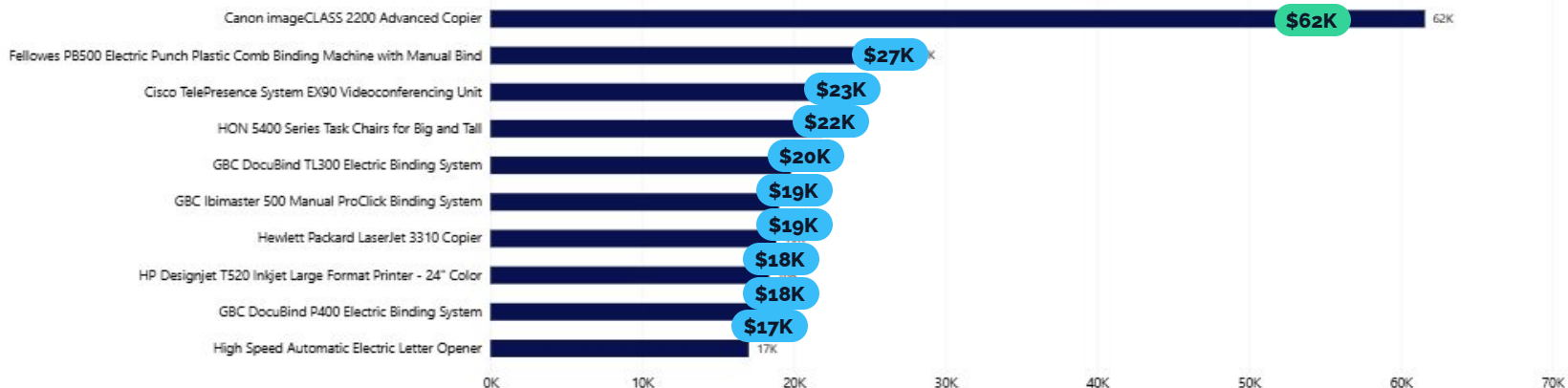
- Copiers generated the highest profit at **\$56K**.
- Phones and Accessories contributed **\$45K** and **\$42K** respectively.
- Tables recorded the largest loss at **-\$18K**.
- Bookcases and Supplies also generated negative profits.
- Profitability varies significantly across sub-categories.

KEY FINDINGS

- Copiers, Phones, and Accessories are the strongest profit contributors.
- Tables represent the biggest profitability challenge.
- Several sub-categories generate strong sales but weak profitability.
- Profit performance is concentrated in a small number of sub-categories.

Top Products Performance Analysis

Top 10 Products by Sales



KEY INSIGHTS

- Canon Copier generated the highest sales (\$62K).
- Technology products dominate the top-selling list.
- Sales are concentrated among a few products.

KEY FINDINGS

- Top products drive a significant share of revenue.
- Product performance helps identify revenue drivers.
- Supports inventory and sales planning.

Strategic Recommendations

QUICK WINS (0–3 Months)

- Review pricing and discount strategies in the Furniture category to improve profitability.
- Focus marketing efforts on the Consumer segment, which generates the highest sales.
- Maintain sufficient inventory for top-selling products to avoid stock shortages.
- Analyze successful sales strategies in the West region and apply them to other regions where appropriate.

LONG-TERM (3–12 Months)

- Continue investing in high-performing categories such as Technology and Office Supplies.
- Use historical sales trends to improve sales forecasting and inventory planning.
- Create a performance dashboard to regularly monitor sales, profit, and key business metrics.
- Review low-profit sub-categories, such as Tables and Bookcases, and identify ways to improve their profitability.

Key Takeaways & Summary

FINANCIAL & REGIONAL HIGHLIGHTS

- Generated **\$2.30M** in Sales and **\$286.4K** in Profit.
- Sales showed an overall upward trend from **2014-2017**.
- **West Region** achieved the highest sales performance.
- **Consumer Segment** contributed the highest revenue.

CATEGORY & PRODUCT DRIVERS

- **Technology** was the most profitable category.
- **Copiers** were the most profitable sub-category, while **Tables** generated losses.
- A small group of top-performing products drove a significant share of revenue.

Thank you

Online Retail Sales & Profitability Analysis

Prepared by: Nirmal
